constexpr reflexpr

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- Audience: SG7 Reflection

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Abstract

Before

```
template <typename T>
T min(const T& a, const T& b) {
  using MetaT = reflexpr(T);
  log() << "min<"
        << get_display_name_v<MetaT>
        << ">(" << a << ", " << b << ") =
  ";
  T result = a < b ? a : b;
  log() << result << std::endl;
  return result;
}
```

After

```
template <typename T>
T min(const T& a, const T& b) {
  constexpr reflect::Type const * metaT =
  reflexpr(T);
  log() << "min<"
        << metaT->get_display_name()
        << ">(" << a << ", " << b << ") = ";
  T result = a < b ? a : b;
  log() << result << std::endl;
  return result;
}
```

The reflection TS working draft (N4766) provide facilities for static reflection that are based on the template metaprogramming paradigm. Recently, however, language features have been proposed that would enable a

more natural syntax for metaprogramming through use of `constexpr` facilities (See: P0598, P0633, P0712, and P0784). This paper explores the impact of these language facilities on reflexpr and considers what a natural-syntax-based reflection library would look like.

Changes

- P0953R1 introduces a new section, Library Design Alternative, which, based on feedback from SG7, presents an approach using type-erased, by-value objects. Second, `typename` was introduced a disambiguator for `unreflexpr` to address ambiguity in some contexts. Third, a discussion was added to explain the motivation of using pointers instead of references. Finally, the reflect namespace is suggested for placement of type_traits functions.

Introduction

From TMP-reflexpr to CXP-reflexpr

Template-metaprogramming-based reflexpr (TMP-reflexpr), which is succinctly described in P0578, includes a `reflexpr` operator that, when applied to C++ syntax, produces a *type* that encodes "meta" information about that syntax. For example, when `reflexpr` is applied to a type, the result can be used to query that type's name and, in the case of a class, list its member variables.

Consider the following implementation of a function `dump` which outputs the name and public data members of an arbitrary type. Note that iteration of the data members involves use of an auxiliary function and variadic templates.

```
template <typename T>
void dumpDataMembers(); // implemented below

template <typename T>
void dump()
    // Output the name and data members of a record (class, struct, union).
    // For example:
    //..
    // struct S {
    //     std::string m_s;
    //     int m_i;
    // };
    //
    // int main {
    //     dump<S>(); // name: S
    //                // members:
    //                //   std::string m_s
    //                //   int m_i
    // }
    //..

{
    using MetaT = reflexpr(T);
    std::cout << "name: " << get_display_name_v<MetaT> << std::endl;
    std::cout << "members:" << std::endl;

    using DataMemberObjectSequence = get_public_data_members_t<MetaT>;
    dumpDataMembers<unpack_sequence<std::tuple, DataMemberObjectSequence>>();
}
```

```
template <>
void dumpDataMembers<std::tuple<>>() { }
    // base case does nothing

template <typename DataMember, typename... DataMembers>
void dumpDataMembers<std::tuple<DataMember, DataMembers...>>()
{
    // Output information about the first data member and recurse for the
    // subsequent data members.
    std::cout
        << "  " << get_display_name_v<get_type_t<DataMember>>
        << " " << get_display_name_v<DataMember>
        << std::endl;
    dumpDataMembers<std::tuple<DataMembers...>>();
}
```

While the above code can be simplified somewhat by use of a template metaprogramming library such as Boost.MPL, this snippet is fairly representative of the complexity and, for most C++ developers, unfamiliarity of the required constructs.

Louis Dionne's Boost.Hana library provides an alternative approach whereby values with specially designed types allow one to write code that has much of the appearance of normal C++ code, but actually accomplishes metaprogramming (See P0425). Instead of having to write a separate function to accomplish iteration, one could use `hana::for_each` from within the `dump` function.

```
template <typename T>
void dump() {
    constexpr auto metaT = reflexpr(T);
    std::cout << "name: " << metaT->get_display_name() << std::endl;
    std::cout << "members:" << std::endl;
    hana::for_each(metaT.get_public_data_members(), [&](auto dataMember) {
        std::cout
            << "  " << dataMember.getType().get_display_name()
            << " " << dataMember.get_display_name()
            << std::endl;
    });
}
```

While this mitigates much of the syntactic complexity of template metaprogramming, the types involved are opaque (they cannot be named) and special library features are required to accomplish tasks such as iteration.

Constexpr-based reflexpr (CXP-reflexpr), the subject of this paper, operates in the Hana-style: instead of `reflexpr` producing a *type*, it produces a *value* that encodes meta information. Unlike Hana-style, however, this value has a non-opaque type and we normally do not need special library features to work with it.

The following snippet illustrates our original example using CXP-reflexpr.

```
template <typename T>
void dump() {
    constexpr reflect::Record const * metaT = reflexpr(T);
    std::cout << "name: " << metaT->get_display_name() << std::endl;
    std::cout << "members:" << std::endl;
    for(RecordMember const * member : metaT->get_public_data_members())
        std::cout
            << "  " << member->get_type()->get_display_name()
            << " " << member->get_name() << std::endl;
}
```

Mix compile-time reflection with runtime values

In the above example, we didn't have a need to mix a runtime value with compile-time meta-information. While this is doable with TMP-reflexpr, additional language facilities are required to accomplish this with CXP-reflexpr.

Consider a function `dumpJson` that outputs an arbitrary `class` or `struct` as a JSON string. Note the use of two new language constructs: `constexpr for`, and `unreflexpr`.

```
void dumpJson(const std::string &s);
    // Output 's' with double-quotes and escaped special characters.

void dumpJson(const int &i);
    // Output 'i' as an int.

template <typename T>
void dumpJson(const T &t) {
    std::cout << "{ ";
    constexpr reflect::Class const * metaT = reflexpr(T);
    bool saw_prev = false;
    constexpr for(RecordMember const * member : metaT->get_public_data_members())
    {
        if(saw_prev) {
            std::cout << ", ";
        }

        std::cout << member->get_name() << "=";

        constexpr reflect::Constant const * pointerToMember = member->get_pointer();
        dumpJson(t.*unreflexpr(pointerToMember));

        saw_prev = true;
    }
    std::cout << " }";
}

struct S {
    std::string m_s;
    int m_i;
};

int main {
    dumpJson( S{"hello", 33} ); // outputs: { m_s="hello", m_i=33 }
}
```

The recursive call warrants some additional explanation

```
dumpJson(t.*unreflexpr(pointerToMember));
```

- `pointerToMember` is "meta" information about a pointer to a member. In the above example, it refers to `&S::m_s` in the first iteration of the loop and `&S::m_i` in the second iteration.
- The `unreflexpr` operator converts this "meta" information about the pointer to member to the pointer to member itself.
- `t.*` is the syntax for accessing a pointer to member.
- `dumpJson` is the recursive call.

The first iteration is decomposed like this:

```
dumpJson(t.*unreflexpr(pointerToMember));
                ↓
dumpJson(t.*&S::m_s);
                ↓
dumpJson(t.m_s);
```

The second iteration is similar:

```
dumpJson(t.*unreflexpr(pointerToMember));
                ↓
dumpJson(t.*&S::m_i);
                ↓
dumpJson(t.m_i);
```

Supporting language constructs

`constexpr for`

This construct essentially unrolls a for loop at compile-time, allowing for the body of the loop to manifest different types at different iterations. This was originally proposed by Andrew Sutton in P0589 for Hana-style programming, but the unrolling semantics work for our purposes as well.

While this language construct isn't strictly necessary for CXP-reflexpr, we think it greatly simplifies code that would otherwise require complex library facilities.

constexpr-time allocators

constexpr-time allocators as described in P0784 (also seen in a more preliminary form in P0597) allow us to make use of `std::vector`, among other data types, at compile time. While these constructs aren't strictly necessary (resizable arrays based on fixed buffer sizes has been demonstrated by Ben Deane and Jason Turner in P0810), they make the data structures used at compile time more uniform with those at runtime.

`unreflexpr`

With TMP-reflexpr, types are easily extracted because we are already in a type context.

```
// 'foo' has a type that is the same as the first field of 'S'.
get_reflected_type_t<
  get_type_t<
    get_element_t<
        0,
        get_public_data_members_t<reflexpr(S)>>>> foo;
```

With CXP-reflexpr, on the other hand, once we have a `Type` object, there isn't a sufficient language facility for going back into type processing

```
constexpr Type const * t = reflexpr(S)->get_public_data_members()[0]->get_type();

// unreflexpr is required to create 'foo' with the type that 't' refers to.
unreflexpr(t) foo;
```

One might think `decltype` would work, but the `Type` returned by reflexpr(X) is generic, and has no specialized member types for `X`; there is nothing `X`-specific to `decltype()` on.

Therefore, the one language-level feature that is critical for feature parity with TMP-reflexpr is `unreflexpr` support. It is required for both types and compile-time values.

Library considerations

A `constexpr reflexpr` facility has several library-level implications and design questions. These are addressed in this section.

Typeful reflection

Some of the initial sketches for constexpr-based reflection had the `reflexpr` operation produce values that are always of the same type. While this has some advantages for implementers and is certainly simpler that each value having its own unique type, we feel that making some use of types will encourage programming that is easier to read and reason about.

- Typeful reflection enables the use of overloading in library development. Consider the simplicity of the following overloaded function when compared to a single function implemented with `constexpr if`.

```
void outputMetaInformation(reflect::Union const *);
void outputMetaInformation(reflect::Class const *);
void outputMetaInformation(reflect::Enum const *);
void outputMetaInformation(reflect::Type const *);
void outputMetaInformation(reflect::TypeAlias const *);
void outputMetaInformation(reflect::Namespace const *);
void outputMetaInformation(reflect::NamespaceAlias const *);
void outputMetaInformation(reflect::RecordMember const *);
void outputMetaInformation(reflect::Variable const *);
void outputMetaInformation(reflect::Enumerator const *);
```

- Typeful reflection is similar to the already accepted and sufficiently motivated concepts-based TMP reflexpr. Types to values is what's concepts are to types.

- Using untyped reflection has many of the same problem as treating all memory as `void*`. Reasonably-strong types help with organization and the long-term maintenance of programs.

We also considered using `std::variant` or variant-like classes as an alternative to reference semantics. Due to the complexity of visitation, the resulting code ended up complex looking, especially for those newer to the language. If a language-level variant with pattern matching support were to be incorporated into C++, then this might be worth revisiting.

References vs. pointers

We were initially tempted to use references instead of pointers to provide a more value-semantic experience. Unfortunately, the inability to put a reference directly in a `std::vector` hinders usability. This design choice would require use of the often confusing `std::reference_wrapper` template in several places.

Downcasting

While reflecting syntax will produce the most-specific type available, the need to go from a general type to a specific type remains. For example, the return type of `Record::get_public_member_types` is `std::vector<Type const*>`. A user may want to "downcast" one of these types into a `Class`, for example.

For this, we provide two cast-related operations in our `Object` class:

```
class Object {
public:
    // ...

    template<typename T>
    constexpr T const * get_as() const;
        // Return the specified view, but constexpr-throw (aka diagnose) in
        // case of invalid accesses.

    template<typename T>
    constexpr bool is_a() const;
        // Returns whether or not this object can be viewed as the specified
        // 'T'.
};
```

These functions allow a user to check if the object can be downcast and to actually perform the operation.

type_traits

The C++ standard library includes several metafunctions that aid in metaprogramming in the `<type_traits>` header. `std::add_const_t` is one such example. While these facilities can be used in the CXP-reflexpr paradigm, it is awkward:

```
Type const * p = /*...*/;
Type const * constP = reflexpr(std::add_const_t<unreflexpr(p)>);
```

We propose that each of these existing metaprogramming functions get a CXP-reflexpr-styled equivalent in the `reflect` namespace.

```
namespace std::reflect {
  constexpr reflect::Type const * add_const(reflect::Type const * t)
  {
    // Not necessarily implemented in this way.
    return reflexpr(std::add_const_t<unreflexpr(t)>);
  }
}
```

Library Design Alternative

While the use of pointers and an inheritance hierarchy for user syntax has benefits, there are two principle problems with this approach. First, as hinted at in P0993r0, the storage and linkage of the pointed-to value raises some implementability concerns that, while likely possible to mitigate, significantly increase the complexity of the approach. Second, we have observed a preference from the committee to use value semantics instead of pointer semantics whenever possible.

P0993r0 advocated for an approach where `reflexpr` would always return values of type `meta::object`. This forces use of concepts in the case of overloading. The following snippet shows the changed signatures of `outputMetaInformation` as described in Typeful reflection:

```
template<reflect::Union u>
void outputMetaInformation();
template<reflect::Class c>
void outputMetaInformation();
```

Note that `reflect::Union` and `reflect::Class` are concepts and not types. `u` and `c`, in this approach, both have type `reflect::object`.

There are three principle drawbacks of this approach. First overloading must always use values passed by a template parameter, even when it would not otherwise be necessary. This may substantially discourage creation and maintenance of reflection code by the large number of C++ developers that would not consider themselves experts in the language. Second, requiring concepts to do basic things goes against the desire to make metaprogramming look just like normal programming. Third, the lack of built-in types may result in the proliferation of programs which use `reflect::object` without concepts and create a maintenance burden.

A third alternative is to use type-erased, by-value objects. This is like the 'pointer' approach in that there is a hierarchy of types, but conversion operators are used instead of casting to base classes. For example, the `RecordMember` type would be,

```
class RecordMember :
{
public:
    constexpr bool is_public() const;
    constexpr bool is_protected() const;
    constexpr bool is_private() const;

    constexpr Record get_type() const;
    operator Named() const;
};
```

, instead of,

```
class RecordMember : public Named
{
public:
    constexpr bool is_public() const;
    constexpr bool is_protected() const;
    constexpr bool is_private() const;

    constexpr Record const * get_type() const;
};
```

. This approach provides the benefits of typeful programming and those of the monotype approach.

Consider the difference between the `get_public_data_members` function of `Record` with the monotype style,

```
class Record : public Type
{
    //...
    constexpr std::vector<meta::object> get_public_data_members() const;
};

// Type provides little information here
std::vector<meta::object> members
  = reflexpr(SomeType).get_public_data_members();

// Alternatively, we assume some kind of terse concepts syntax.
std::vector<meta::RecordMember {}> members
  = reflexpr(SomeType).get_public_data_members();
```

, and the type-erased, by-value style,

```
class Record : public Type
{
    //...
    constexpr std::vector<RecordMember> get_public_data_members() const;
```

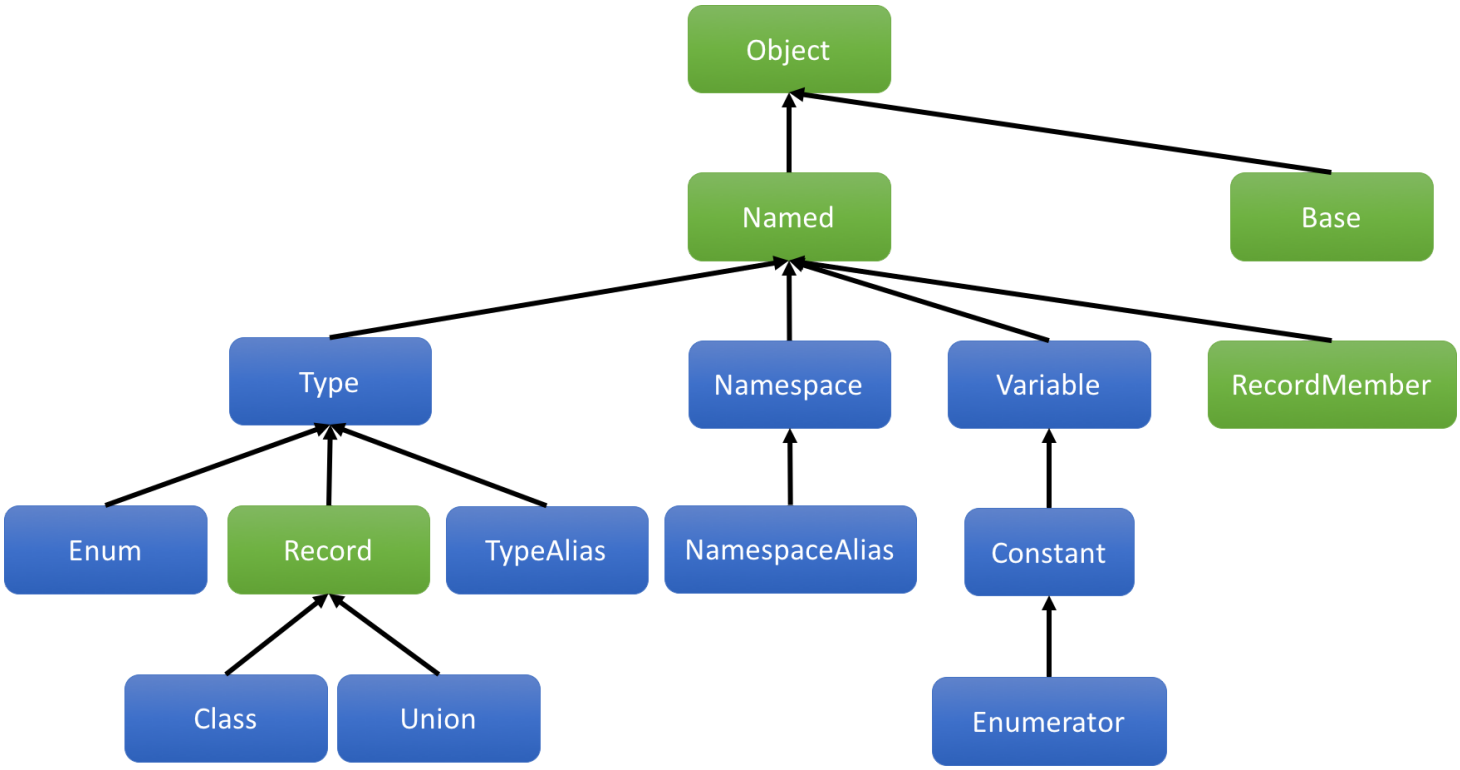
```
};

std::vector<RecordMember> members
  = reflexpr(SomeType).get_public_data_members();
```

. The latter appears at a glance to be typical C++ code.

Datatypes and Operations

CXP-reflexpr provides a rich class hierarchy representing the various attributes that can be reflected. The following diagram illustrates this hierarchy.



Arrows go from derived classes to base classes. The blue classes are those that `reflexpr` directly produces values of. The green classes are those that are intermediate or indirectly available from the other classes.

Classes

What follows is a short synopsis of the class hierarchy described above.

```
namespace reflect {

class Object {
public:
    constexpr std::source_location get_source_location() const;
    constexpr std::string get_source_file_name() const;

    template<typename T>
    constexpr T const * get_as() const;

    template<typename T>
    constexpr bool is_a() const;
};
```

```
class Named : public Object {
    constexpr bool is_anonymous() const;
    constexpr std::string get_name() const;
    constexpr std::string get_display_name() const;
};

class Type : public Named { };

class Record : public Type
{
public:
    constexpr std::vector<RecordMember const*> get_public_data_members() const;
    constexpr std::vector<RecordMember const*> get_accessible_data_members() const;
    constexpr std::vector<RecordMember const*> get_data_members() const;

    constexpr std::vector<Type const*> get_public_member_types() const;
    constexpr std::vector<Type const*> get_accessible_member_types() const;
    constexpr std::vector<Type const*> get_member_types() const;
};

class Union : public Record { };

class Class : public Record
{
public:
    constexpr bool is_struct() const;
    constexpr bool is_class() const;

    constexpr std::vector<Base const*> get_public_bases() const;
    constexpr std::vector<Base const*> get_accessible_bases() const;
    constexpr std::vector<Base const*> get_bases() const;

    constexpr bool is_final() const;
};

class Enum : public Type
{
    constexpr bool is_scoped_enum() const;
    constexpr std::vector<Enumerator const*> get_enumerators() const;
};

class TypeAlias : public Type
{
    constexpr Type const * get_aliased() const;
};

class Variable : public Named
{
    constexpr bool is_constexpr() const;
    constexpr bool is_static() const;

    constexpr Constant const * get_pointer() const;
    constexpr Type const * get_type() const;
};

class Base : public Object
```

```
{
    constexpr Class const * get_class() const;
    constexpr bool is_virtual() const;
    constexpr bool is_public() const;
    constexpr bool is_protected() const;
    constexpr bool is_private() const;
};

class Namespace : public Named
{
    constexpr bool is_global() const;
    constexpr bool is_inline() const;
};

class NamespaceAlias : public Namespace
{
    constexpr Namespace const * get_aliased() const;
};

class RecordMember : public Named
{
    constexpr bool is_public() const;
    constexpr bool is_protected() const;
    constexpr bool is_private() const;

    constexpr Record const * get_type() const;
};

class Constant : public Variable { };

class Enumerator : public Constant { };

} // namespace reflect
```

reflexpr

`reflexpr( texp )` returns values of types according to the first rule that applies:

- If `texp` is a type alias, a `TypeAlias const *` is returned.
- If `texp` is an enum, a `Enum const *` is returned.
- If `texp` is a class, a `Class const *` is returned.
- If `texp` is a union, a `Union const *` is returned.
- If `texp` is any other type, a `Type const *` is returned.
- If `texp` is a namespace alias, a `NamespaceAlias const *` is returned.
- If `texp` is a namespace, a `Namespace const *` is returned.
- If `texp` is an enumerator, a `Enumerator const *` is returned.
- If `texp` is a compile-time constant, a `Constant const *` is returned.
- If `texp` is a variable, a `Variable const *` is returned.

unreflexpr

`unreflexpr(meta)` produces the entities according to the following rules:

- If `meta` has type `Type const*`, the result is the type that `meta` reflects.
- If `meta` has type `Constant const*`, the result is the value that `meta` reflects. In an otherwise ambiguous context, `meta` must have type `Constant const*`.

`unreflexpr typename(meta)` produces the entities according to the following rules:

- `meta` must have type `Type const*`, the result is the type that `meta` reflects.

Conclusion

This paper introduces a facility that allows for compile-time reflection using a natural, constexpr-styled programming by taking advantages of new constexpr language features on the horizon. The result is a much simplified interface that makes both reflection and metaprogramming more accessible and maintainable.

Acknowledgments

None of this would be possible without the continued, pioneering work of Daveed Vandevoorde, Louis Dionne, and Andrew Sutton in improving the experience of constexpr programming.